



PROGRAM NAME B.Tech CSE
YEAR/SEMESTER 4th / 7th
FIRST CONTINUOUS ASSESSMENT, 2024-25

Course Name: Cryptography and Network Security
Time: 1.30 hrs

Course Code: CS 412
Max. Marks: 20

- Note : (i) The question paper contains Three Sections.
(ii) Section-A is compulsory, Section-B and C contains internal choice.

SECTION-A

(MCQS/FILL UP THE BLANKS/VERY SHORT ANSWER QUESTIONS)

1. Attempt ALL parts of the following question:

1 x 5 = 5

[BT-3, CO-1, PO-2]

- (a) Find out the cipher text of below Plaintext using "Playfair Cipher".

Plaintext: ATTACK AFTER NOON

Keyword: PLAYFAIR

- (b) Using the "Rail Fence Cipher" convert the following Plaintext to the Ciphertext. [BT-3, CO-1, PO-2]

Plaintext: ATTACK POSPONED UNTIL TWO AM

Keyword: 4312567

- (c) Differentiate between True Random Number Generator and Pseudorandom number Generator.

[BT-3, CO-2, PO-1]

- (d) What is Decentralized Key Distribution?

[BT-1, CO-2, PO-1]

- (e) What is the difference between Symmetric Key Encryption and Asymmetric Key Encryption.

[BT-1, CO-2, PO-1]

SECTION-B

(SHORT ANSWER QUESTIONS)

2. Attempt any TWO of the following questions:

2.5 x 2 = 5

- (a) What are Security Attacks? Explain the types of Security Attacks.

[BT-1, CO-1, PO-8]

- (b) Differentiated between Block Ciphers and Stream Ciphers.

[BT-3, CO-1, PO-1]

- (c) Explain any one Block Cipher Mode of Operation.

[BT-2, CO-2, PO-2]

- (d) Briefly explain the requirements for a sequence of random numbers.

[BT-2, CO-2, PO-1]

SECTION-C
(LONG ANSWER QUESTIONS)

3. Attempt any TWO of the following questions:

5 x 2 = 10

- (a) Use S-DES to generate the Ciphertext and the Subkeys K1 and K2

[BT-4, CO-1, PO-3]

Plaintext: 1 0 1 1 1 0 1

Key: 1 0 1 0 0 0 0 1 0

Clearly show both the process of Key Generation and Encryption.

Note:

S-box are:

S1 =

1	0	3	2
3	2	1	0
0	2	1	3
3	1	3	2

S2 =

0	1	2	3
2	0	1	3
3	0	1	0
2	1	0	3

- (b) Write the short notes of the following:

[BT-2, CO-1, PO-1]

- i. Steganography
- ii. Rotor Machine

- (c) In reference to Double DES, what is the "Meet-in-the-Middle" Attack, and how can it be overcome using Triple DES.

[BT-3, CO-2, PO-2]

- (d) Explain the process of Key Distribution using KDC with proper diagram.

[BT-2, CO-2, PO-1]